

MDT Æ— graphs, second formulation: pre-registered ledger

The first GNN ledger (GNN_RESEARCH_LEDGER.md) closed all six hypotheses, but its formulation could not have detected a graph advantage: every graph was a kNN re-derivation of the features, datasets were subsampled to 800 nodes, and the harness's Gaussian-euclidean kNN collapses on bag-of-words features (measured ACM feature-graph homophily 0.36 euclidean vs 0.71 cosine). This ledger replaces that formulation. It was written before any GNN result of the new suite existed; the gates and thresholds below are frozen.

Pivotal hypothesis

When at least one view carries relational information not recoverable from the features, graph message passing beats MDT consensus **given the same inputs** (natural adjacencies supplied to MDT as views).

Everything else is an ablation of this claim. The decisive comparison is GNN vs MDT-with-graph-views, never GNN vs feature-only MDT.

Pre-registered gates and thresholds (frozen)

1. **Independence gate.** A dataset enters the primary analysis iff at least one natural view has edge Jaccard $\hat{\propto}$ 0.30 against a cosine feature-kNN graph of matched density (capped at $k=128$). Views with adjusted homophily < 0.05 are flagged heterophilous and analysed as secondary controls.
2. **Minimum useful effect** +0.02 test AML, unchanged from the first ledger.
3. **Verdict rule** (enforced by code, not prose): effects are paired per (dataset, seed), seeds averaged within dataset, 95% t-interval across dataset means. continue iff mean $\hat{\propto}$ +0.02 and lower bound > 0 ; dead_end iff upper bound $< +0.02$; otherwise inconclusive. No verdict from fewer than 5 primary datasets. Secondary: win-rate sign test across datasets.
4. **Model selection is label-free.** One unsupervised criterion selects trajectories, checkpoints, and any per-dataset choice; identical for every arm; labels are read only after a run. Criterion: **modularity of the KMeans(k) partition on the binary union of all input graphs.** Amendment (pre-grid, recorded 2026-07-23): the original criterion was silhouette, which pathologically rewards collapsed diffusion embeddings $\hat{\propto}$ ACM placement measured consensus NMI 0.63 vs silhouette-selected 0.004, with silhouette preferring the degenerate PLP-dominated operator. Replaced before any grid row existed; the broken ACM select rows are the evidence.
5. **No view selection by labels.** The independence screen uses labels to qualify datasets and interpret results, never to configure methods.

Step 1 $\hat{\propto}$ placement against the published record

Full-graph protocol (all nodes, no split), KMeans on the embedding, ACC/macro-F1/NMI/ARI, matching the multiplex clustering literature.

Published anchors (details and caveats in the research notes; two incompatible dataset lineages share names $\hat{\propto}$ we use the DMGI lineage for DBLP-7907 and IMDB-3550, the shared 3025-node ACM):

method	ACM NMI	DBLP-7907 NMI	IMDB-3550 NMI
O2MAC (WWW 2020)	0.692	0.407 (BTGF re-run)	â€”
MCGC (NeurIPS 2021)	0.713	0.551 (BTGF re-run)	â€”
HDMI (WWW 2021)	0.695	0.582 (test split)	0.198 (test split)
BTGF (AAAI 2024)	0.758	0.624	â€”
DMGI (AAAI 2020)	0.687	0.409 (test split)	0.196 (test split)

Caveats: DMGI/HDMI report NMI on test nodes only; O2MAC-lineage numbers are full-graph. Cross-paper re-runs of the same method diverge by up to 30 NMI points (see notes), so placement is against the range, not a single number.

MDT placement arms (all label-free): feat (cosine kNN transition only), graph (natural adjacencies only), all (both), $\tilde{A}$ — fusion mode (consensus = 8 random Dirichlet trajectories, select = best silhouette of 16 sampled + one-hot trajectories), $\tilde{A}$ — embedding dim (n_{classes} , 32). Results: results/graph_mdt/placement.jsonl.

Step 2 â€” independence screen (completed)

results/graph_mdt/independence.jsonl, produced by `python -m experiments.graph_mdt.independence`. Summary:

dataset	view	adj. homophily	kNN adj. homophily	Jaccard	primary?
acm	PAP	0.728	0.563	0.030	yes
acm	PLP	0.455	0.355	0.080	yes
dblp	PAP	0.609	0.429	0.029	yes
dblp	PPrefP	0.540	0.453	0.020	yes
dblp	PTP	0.130	0.300	0.048	yes
dblp	PATAP	0.009	0.300	0.021	heterophilous flag
imdb	MAM	0.167	0.240	0.008	yes
imdb	MDM	0.425	0.297	0.014	yes
cora	A0	0.771	0.488	0.060	yes
citeseer	A0	0.673	0.536	0.114	yes
amazon_ratings	graph	0.140	0.087	0.034	yes
minesweeper	graph	0.009	~0	0.001	heterophilous control
roman_empire	graph	$\hat{A}$ 0.047	0.585	0.002	heterophilous control

Every natural view passes the Jaccard gate: unlike the first formulation, these graphs are not feature re-derivations. Primary datasets: acm, dblp, imdb, cora, citeseer, amazon_ratings (6). Secondary heterophilous controls: minesweeper, roman_empire.

Step 3 “ method grid

arm	question it answers
MLP + feature-space SSL (no graph)	feature ceiling
diffusion/spectral on adjacency only	graph ceiling without learning
MDT consensus incl. graph views	lab baseline, same inputs
GCN and SAGE backbones — GAE and BGRL objectives	the pivotal claim
GNN+MLP distillation	is the graph needed at inference

Decomposition readout: if all $\max(\text{features-only}, \text{graph-only})$ for a method family, fusion failed “ a different conclusion from "GNN failed". DGI stays closed (first ledger, 30/30 losses, objective misalignment).

Grid settings, frozen before any grid row existed: embedding dim 32 for every arm (KMeans still uses $k = \text{class count}$), hidden 128, 150 epochs, checkpoint by the modularity criterion every 15 epochs, encoder BatchNorm after layer 1 (added after the Cora smoke run showed textbook BGRL collapse without it “ recorded as a pre-grid amendment, not tuned on grid results). Fused adjacency rows are capped at their top-64 weights (SAGE-style neighbour sampling, label-free): full-batch passing on the 700+-degree metapath views cost 88 minutes per cell for no design reason. ARPACK non-convergence on a degenerate trajectory skips that trajectory instead of killing the run. All amendments predate the first valid grid row. DBLP only: the same top-64 cap is applied to the MDT transition views (GRAPH_MDT_VIEW_CAP=64) “ PATAP's 28.5M edges make trajectory SVDs intractable otherwise; symmetric with the GNN's capped fused graph and recorded before any DBLP grid row existed.

Step 4 “ datasets

Primary (6): acm, dblp, imdb (DMGI-lineage multiplex), cora, citeseer (intact citation graphs), amazon_ratings. Heterophilous controls (2): minesweeper, roman_empire. Loaders: experiments/graph_mdt/datasets.py, cached under /tmp/multiplex-data/cache. No subsampling: graphs stay intact. ogbn-arxiv deferred until something survives at this scale (CPU-only box).

Step 5 “ evaluation protocol

- Metrics: AMI (primary), NMI, ARI, ACC, macro-F1; KMeans with 10 restarts, 5 seeds; plus spectral clustering on the embedding as a robustness check.
- Inductive protocol (secondary): 70/30 split, embed train, extend to test, assign test points to train-fitted centroids “ same as the first ledger.
- Compute: wall-clock and peak memory per arm; OOS break-even query count.
- Statistics: as in the frozen verdict rule above; heterogeneity across datasets reported (per-dataset effects always shown, never only the pool).
- Verdicts emitted by experiments/graph_mdt code from the intervals; hand-written verdict prose is not admissible.

Commands

```
python -m experiments.graph_mdt.datasets          # convert + cache + smoke
python -m experiments.graph_mdt.independence      # step 2 screen
python -m experiments.graph_mdt.placement        # step 1b placement arms
```

Results (2026-07-24): grid complete, 440/440 rows

Full-graph AMI, mean over 5 seeds (results/graph_mdt/grid.jsonl; verdicts in grid_verdicts.json, emitted mechanically by verdicts.py):

arm	acm	dblp	imdb	cora	citeseer	amazon	minesw.	roman
mdt_consensus_all	0.619	0.550	0.001	0.436	0.381	0.002	0.002	0.287
mdt_select_all	0.522	0.500	0.089	0.419	0.384	0.002	0.000	0.292
mdt_feat	0.509	0.061	0.147	0.285	0.278	0.001	0.000	0.110
mdt_graph	0.022	0.377	0.002	0.160	0.051	0.007	0.000	0.002
gcn_gae	0.441	0.478	0.025	0.380	0.168	0.018	0.050	0.100
gcn_bgrl	0.420	0.520	0.038	0.241	0.070	0.007	0.051	0.109
sage_gae	0.417	0.468	0.044	0.371	0.162	0.020	0.053	0.144
sage_bgrl	0.070	0.060	0.013	0.039	0.018	0.002	-0.000	0.346
mlp_ssl	0.023	0.033	0.012	0.009	0.006	0.001	-0.000	0.350
kmeans_feat	0.379	0.383	0.150	0.082	0.051	0.001	0.000	0.347
distill_mlp	0.424	0.501	0.047	0.377	0.168	0.008	0.000	0.352

Pivotal verdicts (6 primary datasets, frozen rule):

comparison	effect	95% CI	wins	verdict
gcn_gae vs mdt_select_all	-0.068	[-0.152, +0.017]	1/6	dead_end
sage_gae vs mdt_select_all	-0.072	[-0.160, +0.015]	1/6	dead_end
sage_bgrl vs mdt_select_all	-0.286	[-0.492, -0.080]	1/6	dead_end
gcn_bgrl vs mdt_select_all	-0.103	[-0.236, +0.029]	2/6	inconclusive
all four GNN arms vs mdt_consensus_all	-0.08 to -0.30	upper < +0.03	â‰‰ â‰ˆ2/6	dead_end / inconclusive
mdt_select_all vs mdt_graph	+0.216	[+0.022, +0.410]	5/6	continue
distill_mlp vs mdt_select_all	-0.065	[-0.152, +0.022]	2/6	inconclusive

Closure statements:

1. **Pivotal hypothesis rejected in direction.** No GNN arm beats MDT given the same inputs; every mean effect is negative; GNN wins at most 2/6 datasets, and only where the screen showed weak graphs (imdb +0.02-0.04, amazon +0.005-0.02). Formal labels are dead_end where intervals permit and inconclusive elsewhere only because the study cannot exclude a +0.02 GNN advantage “it can and does exclude any large one on these datasets.
2. **Where GNNs lose, they lose to fusion, not features.** GNN arms beat kmeans_feat 5/6 (effect +0.04-0.08, inconclusive); MDT's advantage comes from consensus over feature+graph views (fusion readout: continue).
3. **Controls behaved as designed.** roman_empire (anti-correlated graph): message passing hurts every GNN (-0.14 to -0.19 vs MDT; feature-only arms win at 0.35). minesweeper (structure-only): GNNs +0.05 over everything “the only regime with a graph-specific gain, and it is small.
4. **Trajectory selection: consensus beats the silhouette/select recipe** on acm (+0.10) and dblp (+0.05); the silhouette criterion itself was shown to prefer collapsed embeddings (amendment above). Actionable for the lab independent of any GNN question.
5. **amazon_ratings carries no recoverable cluster signal for any method** (max AMI 0.02) “retained in the primary analysis per pre-registration.
6. **Placement (step 1) headline.** Zero-training MDT consensus with graph views reaches NMI 0.63 on ACM and 0.59 on DBLP-7907 vs published GNN clustering 0.69-0.76 and 0.41-0.62 respectively; on DBLP it beats HDML and every pre-2024 method. The field's GNN-over-spectral margin on these benchmarks is small once the spectral method is given the same graphs.

Enhancement round E1/E3 (pre-registered 2026-07-24, before any E-row)

The grid located MDT's failures: noise views poison uniform consensus (imdb, acm-PLP), and diffusion cannot use heterophilous structure (minesweeper is the only regime with a GNN gain, +0.05). Two heuristic arms test the cheap fix first; a GNN version is licensed only if its heuristic shows the effect exists.

- **E1 mdt_gated** “per-node view gating. Gate of view v at node i is the cosine between node i 's raw features and its view- v neighbour average ($P_v X$, self-loop-free); gates are clipped at 0 and normalised per node; per-step Dirichlet weights modulate the gates and the mixture stays row-stochastic. Consensus over 8 trajectories as in `mdt_consensus_all`. Global learned weights stay closed; this tests sample-dependent gating, the reopening clause of both ledgers.
- **E3 mdt_a2** “two-hop derived views. For every natural adjacency A , add binarised A^2 (diagonal removed, rows capped at top-64) as an extra view to the `mdt_consensus_all` recipe. Tests whether “same-role” 2-hop homophily captures the heterophilous signal without any GNN.

Frozen readouts (same verdict rule, same 6 primary datasets): 1. `mdt_gated` vs `mdt_consensus_all` “E1 primary. 2. `mdt_a2` vs `mdt_consensus_all` “E3 primary. 3. Rescue check: `mdt_gated` on imdb “the 0.147 feat-only AMI means gating defeats the noise-view failure; licenses a learned (GNN) gate. 4. Control check: `mdt_a2` vs `gcn_gae` on minesweeper “if A^2 closes the +0.05 gap, the heterophily argument for GNNs dies too. 5. A combined `gated+A^2` arm runs only if both individual arms clear their primary readout.

Amendment before the first recorded E-row: the acm/imdb seed-0 smoke showed the frozen cosine gates are too soft to defeat imdb's noise views (feature view $\hat{\omega}^0.4$ weight, gated AMI 0.001). A sharpened variant `mdt_gated_sharp` (gates⁴, renormalised; $\hat{I}^2=4$ fixed a priori, no sweep) runs as a **secondary** arm. Because the smoke touched imdb " the rescue dataset " the sharp variant cannot claim the pre-registered rescue readout; only `mdt_gated` can. The sharp arm is exploratory and any positive result needs fresh validation.

E1/E3 results (2026-07-24, 120/120 rows)

AMI, mean over 5 seeds:

arm	acm	dblp	imdb	cora	citeseer	amazon	minesw.	roman
mdt_consensus_all (ref)	0.619	0.550	0.001	0.436	0.381	0.002	0.002	0.287
mdt_gated	0.624	0.150	0.001	0.379	0.324	0.002	0.008	0.318
mdt_gated_sharp (expl.)	0.685	0.032	0.001	0.342	0.220	0.002	-0.000	0.222
mdt_a2	0.596	0.414	0.001	0.454	0.322	0.003	0.002	0.173

Verdicts (frozen rule): `mdt_gated` vs consensus -0.085 [-0.250, +0.080], 1/6 wins, inconclusive; `mdt_a2` vs consensus -0.033 [-0.093, +0.027], 2/6, inconclusive; `mdt_a2` vs `gcn_gae` on minesweeper -0.048 " **A² does not close the GNN's structure-only gap**, the one graph-specific gain stands.

Closure:

1. **E1 heuristic fails its primary readout and the imdb rescue** (gated imdb 0.001 vs 0.147 feat-only). Under the frozen licensing rule, a learned (GNN) gate is **not licensed** by this round.
2. **The failure is diagnostic, not random.** The gate anchors on feature consistency, so it helps exactly where features are strong (acm +0.07 AMI / +0.21 ACC for the sharp variant: ACC 0.886, NMI 0.685 " O2MAC-level from a zero-training spectral method) and destroys where features are weak (dblp 0.55 " 0.03). Per-node gating has real leverage; the signal must estimate view quality without anchoring on any single view. A future round may pre-register e.g. cross-view neighbourhood agreement as the gate signal; it starts as a new hypothesis, not a continuation of this one.
3. **E3 dead as an MDT enhancement** and as an explanation of the GNN's minesweeper edge. The +0.05 structure-only GNN gain on minesweeper is the single surviving graph-specific result in the whole programme; it is small and lives on a synthetic control.
4. The exploratory sharp arm's acm result (ACC 0.886) was obtained after the smoke touched acm and carries no pre-registered status; treating it as a claim requires fresh validation.

Sharp-gate validation on untouched data (pre-registered 2026-07-24)

The exploratory `mdt_gated_sharp` acm result (ACC 0.886 / NMI 0.685) gets one validation shot on the DMGI `amazon` multiplex pickle " never loaded by any prior run of this programme.

Frozen protocol: identical recipes, 5 seeds, MDT-family arms plus kmeans_feat reference only (GNN arms skipped; no GNN claim is under validation). Readouts, frozen before the first row:

1. mdt_gated_sharp vs mdt_consensus_all on amazon “ replication check. The gate anchors on feature consistency, so the prediction is conditional: it should help only if amazon's feature view is strong (mdt_feat near the best arm) and hurt otherwise.
2. mdt_gated (primary formula) same comparison, same prediction.
3. No further amendment of gate formulas after this run; a third gate design requires a new hypothesis section.

Validation results (2026-07-24, 40/40 rows)

AMI, mean over 5 seeds: kmeans_feat 0.048, mdt_feat 0.021, mdt_gated_sharp 0.014, mdt_gated 0.011, mdt_consensus_all 0.001, mdt_select_all/mdt_a2/mdt_graph $\hat{\%}^0.000$.

1. **The amazon dataset is uninformative for the replication check:** no method exceeds 0.05 AMI (like amazon_ratings). The exploratory acm sharp-gate result (ACC 0.886) therefore remains **unvalidated** “ it gains no support and suffers no refutation here. It must not be presented as more than exploratory.
2. **The conditional gate model is refined, not confirmed.** Gates beat consensus on amazon (+0.013/+0.010) despite weak features, because the graph views are pure noise (mdt_graph 0.000) and the gate correctly leans toward the only informative view. Full-programme pattern: the feature-anchored gate helps when features are at least as informative as the graphs (acm, amazon, roman_empire) and hurts when graphs dominate (dblp, cora, citeseer). That is exactly what a feature-anchored signal must do; it estimates which view agrees with features, not which view is good.
3. Gate-formula work is closed per readout 3. A view-quality signal that does not anchor on any single view (e.g. cross-view neighbourhood agreement) remains the licensed future hypothesis, as a new pre-registration.

Kill criteria

- Placement shows MDT-all inside the published GNN range on ACM/DBLP/IMDB “ the field's GNN gain over spectral consensus is smaller than reported; proceed to the controlled grid with that as the headline.
- Placement shows MDT-all far below the published range “ MDT fusion is the bottleneck; fix or close before spending GNN compute.
- Grid: pivotal comparison dead by the frozen rule on “¥5 primary datasets “ close the branch; the only reopening is genuinely relational side information of a different kind (sample-dependent gating already excluded).